

Deep Learning Approaches to Universal Document Parsing

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1. Introduction

Document ingestion platforms face challenges when multi-column typography splits visual sentences. Traditional horizontal text sorting merges lines from adjacent columns incorrectly into single phrases.

2. Related Work

Heuristic layout models detect vertical gutters to segment left and right text streams sequentially. Our approach preserves reading order cleanly.

3. Experimental Methodology

We evaluate candidate document extractors against 13 multi-format suites covering tables, equations, scans, and presentation shapes.

4. Quantitative Results

The cascading architecture achieves 100% flow integrity across multi-column test runs. Throughput exceeds 10 pages per second.

5. Conclusion

Gutter-aware parsing guarantees fidelity.